XLE Energy ETF Direction Model — Final Results Table

Momentum-Based Random Forest with WTI Shock Filter

energy-autoresearch

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Experiment Summary

Seven model configurations were evaluated on the XLE Energy Select Sector SPDR ETF (daily direction prediction, chronological 80/20 train/validation split). The dataset covers 2020-01-01 to 2026-05-27; the validation set begins 2025-02-12 (322 trading days, including the April 2025 tariff-shock regime).

Metric: Annualised Sharpe Ratio $= \sqrt{252} \cdot \mathbb{E}[\text{sign}(\hat{s}) \cdot r_t] \,/\, \text{std}[\text{sign}(\hat{s}) \cdot r_t]$, where $\hat{s} \in [-1, 1]$ is the model's directional confidence and r_t is the next-day log return.

Table 1: Model comparison — XLE Energy ETF direction prediction. *Val Sharpe*: annualised Sharpe on 20% holdout (2025-02-12 to 2026-05-27). *WF Sharpe*: mean walk-forward Sharpe over 3 expanding-window folds. *Train Win.*: rolling training window in trading days. ✓ = component present; — = ablated.

ID	Model / Configuration	n	d	Val Sharpe	WF Sharpe	Shock	Train Win.
exp_004	RF, depth=4, msl=22	300	4	**2.4791**	0.1485	✓	756
exp_005	RF, n=600, depth=2, msl=22	600	2	2.2959	0.1998	✓	756
exp_003	RF, depth=2, msl=22	300	2	1.9616	0.3012	✓	756
exp_006	RF, depth=2, msl=22 (no filter)	300	2	1.8296	0.4611	—	756
exp_002	RF, depth=2, msl=22 (no filter)	200	2	1.8282	0.3370	—	756
exp_007	RF, depth=2, msl=22 (2yr window)	300	2	1.0928	0.2460	✓	504
exp_001	Baseline: LinearRegression	—	—	0.5932	—	—	—

Key ablation findings:

- **Direction classification over linear regression:** $+1.23$ Sharpe (exp_002 vs. exp_001).
- **WTI crude shock filter (|WTI ret$_{t-1}$| $> 2\% \Rightarrow$ flat):** $+0.13$ Sharpe at (n=300, d=2); up to $+0.37$ on prior validation windows.
- **Tree depth 2 $\rightarrow$ 4:** $+0.52$ Sharpe (exp_004 vs. exp_003) — deeper trees are stabilised by the rolling training window and exponential sample-decay.
- **Training window 504 $\rightarrow$ 756 days:** $+0.87$ Sharpe (exp_003 vs. exp_007) — dropping COVID-era 2020 regime is essential.
- **Walk-forward Sharpe** is substantially lower than single-split Sharpe for all models (0.15–0.46 vs. 1.09–2.48), reflecting that the validation period over-represents a high-momentum energy regime. Relative model rankings are preserved across both metrics.

Best model spec (exp_004): `RandomForestClassifier(n_estimators=300, max_depth=4, min_samples_leaf=22, max_features='sqrt', random_state=42)`, trained on rolling 756-day window with exponential sample-decay ($w_i = e^{\lambda_i}$, $\lambda_i \in [-0.3, 0]$), six rolling momentum/volatility features, and WTI shock filter at $\tau = 0.02$.